Corollary A1. *For any single covector v_μ the $P^{(q)}$ projector annihilates the two rank-one seeds: $P^{(q)}[\delta^A{}_{[B}v_{C]}] = 0 = P^{(q)}[\epsilon^A{}_{BCD}v^D]$.*

Appendix B.5. Consequences for the C1 Ansatz

Applying Proposition A3 to the most general *linear* ansatz with one derivative (Section 4.1),

$$T^A{}_{BC} = A\,\delta^A{}_{[B}\partial_{C]}\epsilon + B\,\epsilon^A{}_{BCD}\partial^D\epsilon + \big(\mathcal{P}\cdot\partial\epsilon\big)^A{}_{BC}, \tag{A27}$$

shows that the attempted traceless piece $\big(\mathcal{P}\cdot\partial\epsilon\big)^A{}_{BC}$ necessarily vanishes: it is a linear, single-vector construct and is thus killed by $P^{(q)}$ (Corollary A1). The ansatz collapses to

$$T^A{}_{BC} = A\,\delta^A{}_{[B}\partial_{C]}\epsilon + B\,\epsilon^A{}_{BCD}\partial^D\epsilon, \tag{A28}$$

recovering the corresponding equation in the main text. The scalar projector $\Pi_{\rm PT}[\cdot]$ removes PT-odd *scalars* built from the axial seed (Theorem 1), and the Palatini connection equation then sets $B = 0$ while fixing $A = 2\eta$ under the trace lock $T_\mu = 3\eta\,\partial_\mu\epsilon$ (Section 4.1). This yields the uniqueness map $T^A{}_{BC} = 2\eta\,\delta^A{}_{[B}\partial_{C]}\epsilon$ quoted in Theorem 2.

Appendix B.6. Consistency Check with the Quadratic Invariant

With the normalization (A21) and the C1 map (so $S^\mu = 0$ and $q_{\lambda\mu\nu} = 0$),

$$\begin{aligned}\Pi_{\rm PT}\Big[T^A{}_{BC}T_A{}^{BC}\Big] &= \frac{2}{3}\,T_\mu T^\mu = \frac{2}{3}\,(3\eta)^2\,\Sigma_\epsilon = 6\,\eta^2\,\Sigma_\epsilon,\\ I_T \equiv -\,\Pi_{\rm PT}\Big[T^A{}_{BC}T_A{}^{BC}\Big] &= -6\,\eta^2\,\Sigma_\epsilon,\end{aligned} \tag{A29}$$

in agreement with the main-text definition (12) and the post-lock collapse in Section 3.2.

Appendix C. Three-Chain Reductions and Improvement Currents (σ_ϵ Scheme)

Scope (σ_ϵ scheme and naming).

This appendix provides the paper-checkable reductions behind Section 5 under the *sign-compensated* convention $\mathrm{sgn}(\Sigma_\epsilon)\,I_T \equiv \mathrm{sgn}(\Sigma_\epsilon)I_T = -6\eta^2|\Sigma_\epsilon|$ and the rank-one determinant route naming (formerly "DBI"-type; not Born–Infeld gravity): (i) a rank-one determinant route built out of the canonical traceless matrix $\mathcal{T}^\mu{}_\nu$, (ii) a closed-metric rank-one deformation, and (iii) the PT-even CS/Nieh–Yan shadow. At quadratic order, *each route* reduces in the bulk to the same invariant line

$$\boxed{\begin{aligned}&\delta^2\mathcal{L}_X = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\,I_T + \nabla_\mu J^\mu_X,\\ &\quad A_\star = \lambda^2/8, \qquad X \in \{\mathit{rank-one\,determinant\,route}, \mathrm{CM}, \mathrm{CS}^+\}.\end{aligned}} \tag{A30}$$

with improvements $\nabla_\mu J^\mu_X$ differing by boundary choices (A4/A5). Closed representatives for J^μ are given on FRW/weak-field backgrounds.

Notation and key relation.

We use the preamble shorthands $\Sigma_\epsilon \equiv \Pi_{\rm PT}\big[(\partial\epsilon)^2\big]$, $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$, $\hat{n}_\mu \equiv \dfrac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}}$, $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu\hat{n}_\nu - \frac{1}{4}\delta^\mu{}_\nu\Big)$, $\tau \equiv \hat{n}^\mu T_\mu = 3\,\eta\,\mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}$ so that $\mathrm{Tr}\mathcal{T} = 0$, $\mathrm{Tr}(\mathcal{T}^2) = \frac{3}{4}$, and (after C1)

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon)I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\,I_T \equiv \mathrm{sgn}(\Sigma_\epsilon)I_T = -6\eta^2|\Sigma_\epsilon|.$$